

Integrating Worldwide University Data from Wikidata, ROR and DEQAR

Project Abstract – Web Data Integration, HWS 2026

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1 Use Case

Students, researchers and other stakeholders who need information about universities worldwide face a fragmented landscape: registries, knowledge bases and quality assurance databases differ in format, definitions and coverage, which makes institutions hard to compare. The Research Organization Registry (ROR) has reliable names and locations, but few descriptive facts. Wikidata adds founding years, student numbers, institution types and memberships, but it is incomplete. The Database of External Quality Assurance Results (DEQAR) of the European Quality Assurance Register for Higher Education (EQAR) records the official name of an institution, its mergers and closures and its external quality assurance reports, but 89% of its institutions lie in the European Higher Education Area.

We integrate these three sources into one coherent dataset that describes every university exactly once. Students can look up where a university is located, how old and how large it is and whether it has been quality-assured; analysts, policy makers and university administrators can compare institutions without consolidating sources by hand.

What we count as a university. A university in this project is a higher education institution that awards academic degrees and is an organisation of its own. The sources draw this line differently, so two kinds of records need care. (i) *Parts of a university*: the type *education* of ROR also holds colleges, institutes and branches; 2,786 of its records name a parent organisation, e.g. “Islamic Azad University South Tehran Branch”. DEQAR lists faculties, schools and departments of Spanish universities (1,275 records with a parent institution, such as “Faculty of Law”) and single sites of French schools. (ii) *Predecessors*: DEQAR keeps 916 closed institutions and records mergers, absorptions and split-offs for 1,054 institutions, e.g. Paris-Sorbonne University, closed in 2018 and succeeded by Sorbonne Université. A part is never merged with the university it belongs to, and a predecessor is a different entity than its successor, whereas a former name of the same institution is one of its alternative names. Such records stay in the input data; they are the hard non-matches of our gold standard.

2 Datasets: Schema and Basic Profile

Table 1 profiles the three datasets. Each contains a single class, *University*, so the number of records per class is the number of entities in the table.

Data sources and preprocessing. Our Python scripts retrieved Wikidata and ROR on 2 October 2026 and DEQAR on 9 October 2026 and store each source as one CSV file with one row per university. Attributes with several values are kept as lists, and website links that differ only in **http/https** or **www** are merged.

- *Wikidata*, <https://query.wikidata.org/sparql>: all items that are an instance of *university* (Q3918) or of one of its subclasses. Identifiers of countries, places and types were replaced by English labels; per university we kept the earliest founding year, the preferred or most recent student number and the city of its headquarters or location.
- *ROR*, data dump v2.13 of 22 September 2026, <https://doi.org/10.5281/zenodo.22902037>: of its 141,528 records we kept the 30,131 of type *education* that are not withdrawn.
- *DEQAR*, <https://www.eqar.eu/qa-results/download-data-sets/>: the list of higher education institutions that EQAR publishes as a CSV file and refreshes every night. It is based on records of the European Tertiary Education Register (ETER) and on entries by the quality assurance agencies registered with EQAR. We merged the rows that repeat an institution once per location, kept the year of the founding and closure dates and, where the file gives no country, took the country from the code that starts the ETER identifier.

Table 1. Datasets

Dataset	Source (*)	Format	# of entities	# of attributes	List of attributes (**)
Wikidata	query.wikidata.org/sparql	CSV	22,697	16	Wikidata id, name, alt. names (MV), acronyms (MV), country, country code, city, website, founding year, latitude (MV), longitude (MV), students (MV), types, member of (MV), ROR id (MV), Wikipedia title (MV)
ROR	doi.org/10.5281/zenodo.22902037	CSV	30,131	18	ROR id, name, alt. names, acronyms (MV), country, country code, region, city, website, domains (MV), founding year, latitude, longitude, types, status, parent organisation (MV), Wikidata id, Wikipedia title (MV)
DEQAR	eqar.eu/qa-results/download-data-sets	CSV	8,979	13	DEQAR id, name, alt. names (MV), country, city, website, founding year (MV), closure year (MV), parent institutions (MV), history (MV), QA reports (MV), last QA report (MV), ROR id (MV)

(*) Retrieved with our own Python scripts from the address shown: Wikidata by SPARQL queries and ROR as a data dump on 2 October 2026, DEQAR as a CSV download on 9 October 2026; see “Data sources and preprocessing”. (**) (MV) = more than 30% missing values. The identifiers and the links to other sources serve only as gold standard for the matching.

Missing values. Over all attributes, 32% of the values are missing in Wikidata, 17% in ROR and 39% in DEQAR. In DEQAR this is partly by nature: a closure year and a history exist only for institutions that were closed or merged. The attributes we match on are well filled: the name is never missing, and the country is missing in 3% of the Wikidata, 1% of the DEQAR and none of the ROR records. The sources fill each other’s gaps: the founding year is missing in 19% of the Wikidata and 41% of the DEQAR records but in only 11% of the ROR records, and the coordinates missing in 31% of the Wikidata records are complete in ROR.

3 Integrated Schema and Overlap with Input Schemata

The integrated schema has one class, *University*, with 21 attributes (Table 2): 11 are contained in at least two datasets, five other than the name in all three (for fusion by voting), and seven are lists. No single attribute serves as a key: the source identifiers are not part of the schema, and the name alone is ambiguous (in ROR, “National University” exists in four countries). A university is identified by the combination of name and country, unique for 99.1% of the Wikidata, 99.8% of the ROR and 94.5% of the DEQAR records; in DEQAR the Spanish faculties share names such as “Faculty of Law”, and with the city and the parent institution 99.8% of its records are unique.

Table 2. Attribute Intersection with Integrated Schema

Attribute name	Attribute type	Datasets in which the attribute is found
name	string	Wikidata, ROR, DEQAR
alternative names	string/list	Wikidata, ROR, DEQAR
country	string	Wikidata, ROR, DEQAR
city	string	Wikidata, ROR, DEQAR
website	string	Wikidata, ROR, DEQAR
founding year	date	Wikidata, ROR, DEQAR
acronyms	string/list	Wikidata, ROR
country code	string	Wikidata, ROR
latitude	decimal	Wikidata, ROR
longitude	decimal	Wikidata, ROR
parent organisation	string/list	ROR, DEQAR
region	string	ROR
status	string	ROR
internet domains	string/list	ROR
institution types	string/list	Wikidata
number of students	integer	Wikidata
memberships	string/list	Wikidata
closure year	date	DEQAR
history (mergers and split-offs)	string/list	DEQAR
number of quality assurance reports	integer	DEQAR
date of last quality assurance report	date	DEQAR

The mapping is not one-to-one everywhere: ROR names a parent organisation by its identifier, which we replace by the name, and the *types* of ROR only hold registry categories such as *education* or *funder*, so they are not mapped to the institution types.

4 Entities Contained in Multiple Datasets

Estimate based on small samples. We drew two random samples of 100 records. Of 100 Wikidata universities, 56 carry a link to a ROR record, which extrapolates to about 12,700 entities in at least two datasets. Of 100 DEQAR institutions, 41 carry a link to a ROR record; we checked the other 59 by hand and found 19 more in ROR or Wikidata, so 60 are contained in another dataset, which extrapolates to about 5,400. Both estimates are far above the required 1,000.

Check on the complete files. Wikidata and ROR reference each other through ROR and Wikidata identifiers, and DEQAR carries ROR identifiers. Following these links groups the 61,807 records into 45,081 distinct institutions, 14,372 of them in at least two sources and 2,183 in all three (Table 3); an exact comparison of names and countries raises these lower bounds to 14,658 and 2,627. The identifiers are removed from the matching input, and names alone do not suffice: 25% of the linked Wikidata–ROR and 52% of the linked DEQAR–ROR pairs differ in name even after normalisation.

Table 3. Project requirements and how the project meets them

Requirement	Our project
1. 3 different data sets	3: Wikidata, ROR, DEQAR
2. at least 2,500 entities in total (good: > 10,000 but < 100,000)	about 45,000 (from 61,807 records)
3. at least 1,000 entities contained in at least two datasets	about 14,400
4. at least 8 attributes; identifiable by at least two attributes	21; name + country
5. at least 5 attributes in at least two datasets; some in three	11; 5 other than name in three
6. at least one list attribute	7, e.g. alternative names, history

Gold standard. The links give us matching pairs, but they are incomplete: 19 of the 60 DEQAR sample institutions that are contained in another dataset carry no link. We therefore do not derive non-matches from missing links; we check pairs that look alike but are not linked by hand and label them as match or non-match. We add corner cases by hand: a faculty, site or branch against its university (16 of the 59 unlinked sample records are such parts), a closed predecessor against its successor (2 of the 59), and universities that share a name in different countries.

Distinct sources. Compared with our first abstract we dropped THE and DBpedia. DBpedia is extracted from the English Wikipedia, on which Wikidata partly draws as well: 63% of the linked Wikidata–DBpedia pairs have coordinates that are identical to about ten metres. The remaining sources have different origins: Wikidata is edited by volunteers, ROR is curated by its own team, and DEQAR is based on ETER and on the entries of quality assurance agencies. On linked records the founding year of DEQAR equals that of ROR in only 73% and that of Wikidata in 74% of the cases. DEQAR is the smallest dataset, but about 60% of its institutions are contained in another dataset, and it shares five attributes besides the name with both.